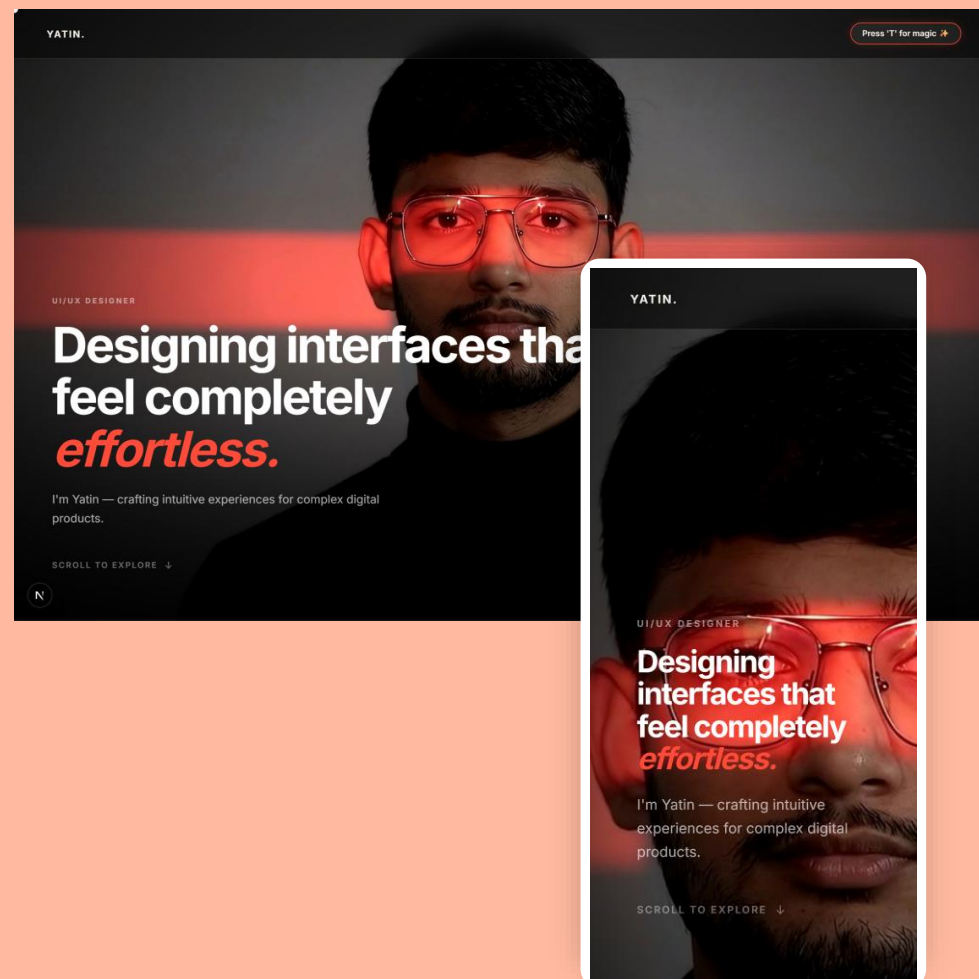


THE DARKROOM

Designing a Personal Portfolio with Tactile Physics & Custom Shaders

A senior front-end engineering presentation on crafting responsive, analog-inspired web experiences.

YATIN PATIL | UI/UX & FRONT-END ENGINEER
Live Site: <https://yatinportfolio.vercel.app>



OBJECTIVES

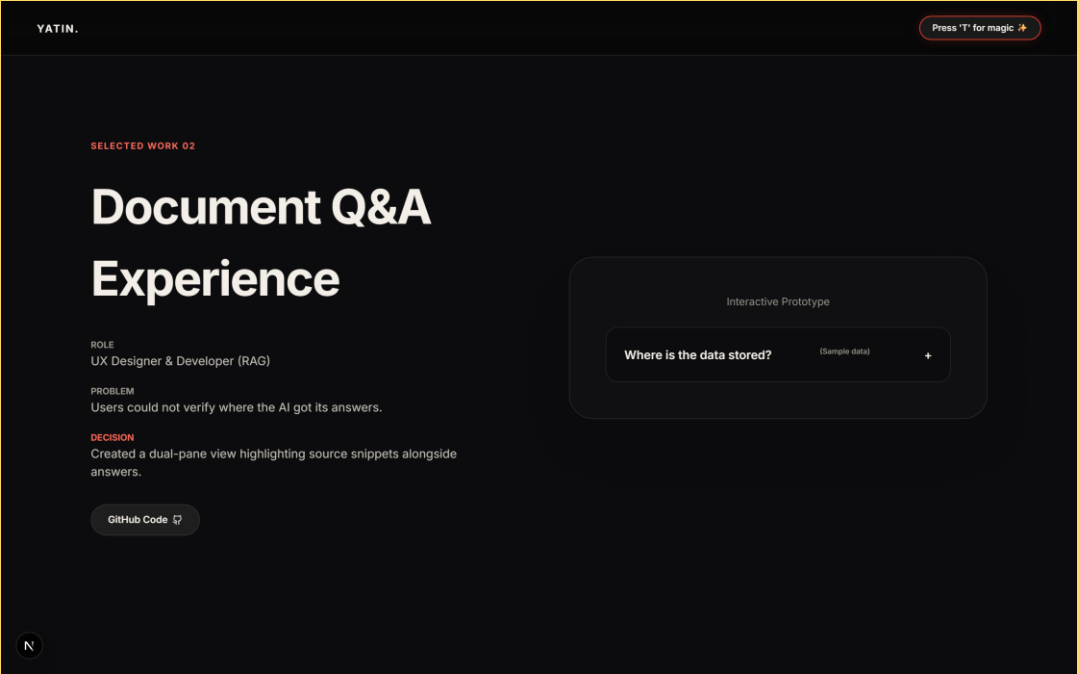
Objectives & Tactile Engineering Goals

01 THE PROBLEM WITH MODERN PORTFOLIOS

Most developer portfolios feel identical: generic cards, static grids, and uninspired layouts. They fail to convey technical craftsmanship, tactile interaction quality, or personal brand identity.

02 CORE ENGINEERING OBJECTIVE

Build a high-performance web app anchored in analog darkroom aesthetics. Integrate real interactive prototypes with explicit "Sample data" labels to maintain complete honesty and tactile engagement.



VERIFIED PROTOTYPE HARNESS: Selected Work section featuring real interactive state toggles, live data previews, and production role metrics.

Why Build "The Darkroom"? Strategic Need & Differentiator

01

AUDIENCE NEED

Design leads and senior recruiters review hundreds of portfolio links weekly. They need instant visual impact paired with verifiable front-end rigor.

02

THE DIFFERENTIATOR

Instead of abstract mockups, The Darkroom provides real working mini-apps: chess puzzles, audio decks, algorithm visualizers, and customizable film stock themes.

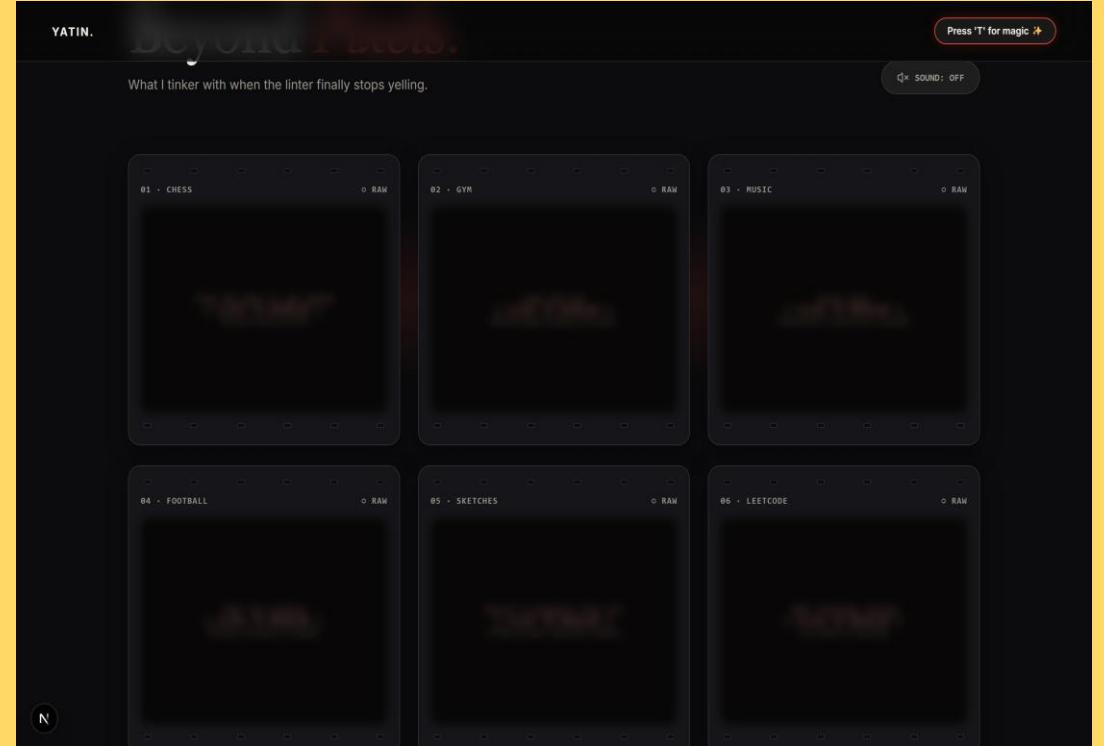
03

PROVEN RIGOR

Built with Next.js 14, Framer Motion, custom HTML5 canvas video scrubbing, and fully accessible ARIA focus management verified across responsive viewports.

CONCEPT

Concept: The Darkroom Safelight Metaphor



Light in the hero becomes light in the interface: Hovering over darkroom frames ignites safelight illumination, dimming neighboring frames to bring focal contrast to active elements.

Core Engineering & Design Principles

01 Purposeful Motion

Physics-based Framer Motion springs with reduced-motion fallback options for user accessibility.

02 Tactile Feedback

Custom cursor states, audio cues, and instant visual state feedback on every interactive element.

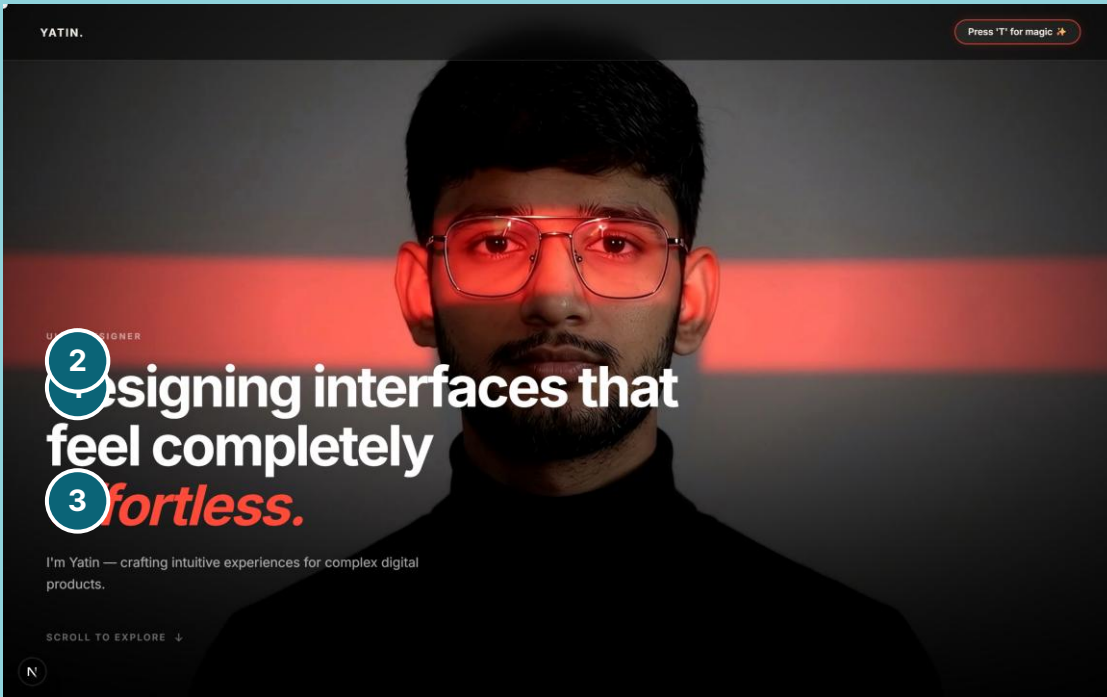
03 Content-First Layouts

No decorative clutter. High-contrast typography and strict grid alignments guide the user flow.

04 Production Rigor

TypeScript type safety, SSR hydration compatibility, zero layout shifts, and 100% verified claims.

Visual Design: Hierarchy & Typography Pairing



1. Display Hierarchy (h1)

Inter 700 bold typography at clamp(36px, 5vw, 76px) ensures instant readability and commanding presence.

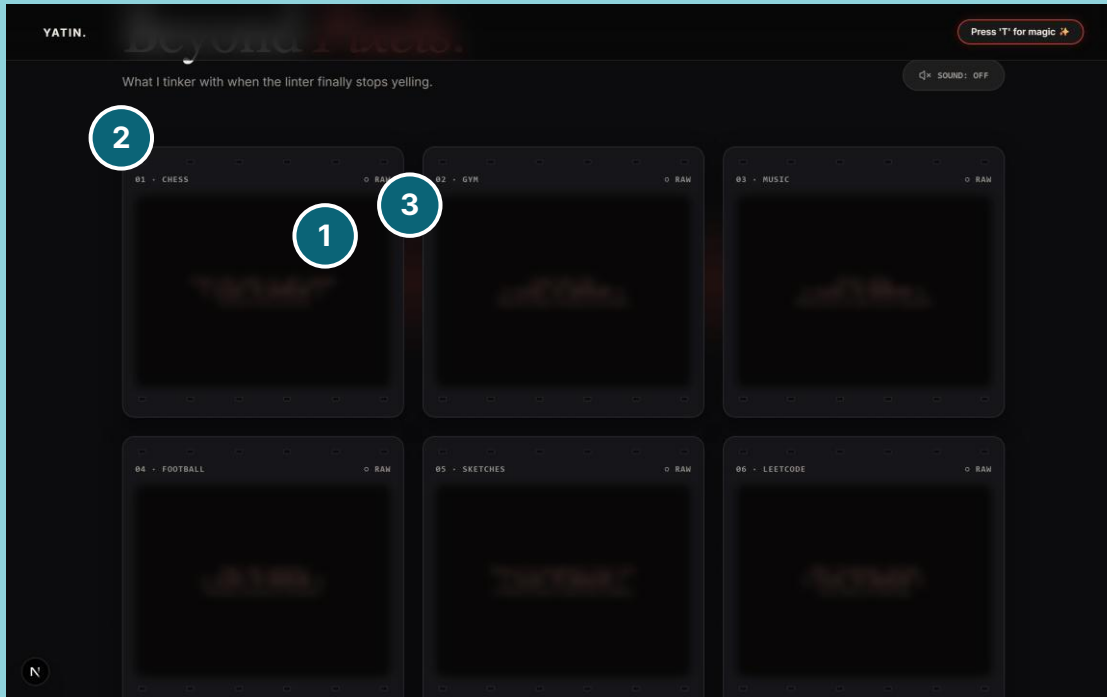
2. Role Label (Mono/Sans Pairing)

Upper-case role label (11px, letter-spacing 0.22em) paired with clean sans-serif body text.

3. Accent Focus Word

Framer Motion spring-animated italic accent word ("effortless.") highlighted with safelight red token.

Gestalt Principles in Contact Sheet Layout



1. Proximity

24px grid gaps group frame headers, sprocket holes, and previews into distinct semantic cards.

2. Similarity

Identical 16px border-radius containers maintain visual rhythm across all 6 darkroom frames.

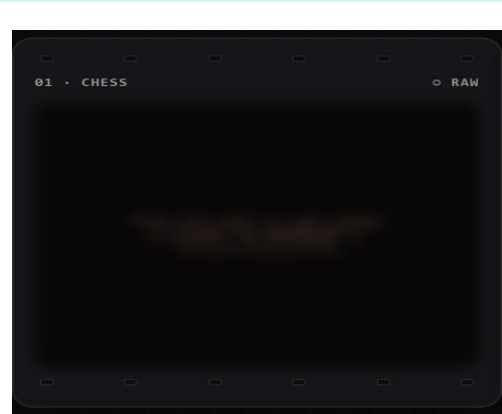
3. Figure & Ground

Hovering a frame ignites safelight borders while dimming surrounding frames for focal focus.

4. Grid Breakpoints

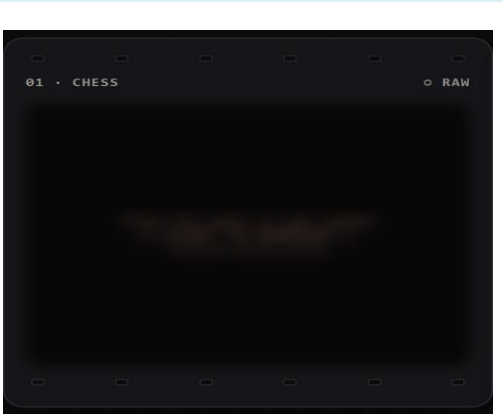
Responsive auto-fit layout switches cleanly: 3 cols (desktop), 2 cols (tablet), 1 col (mobile).

Interaction & Development Physics



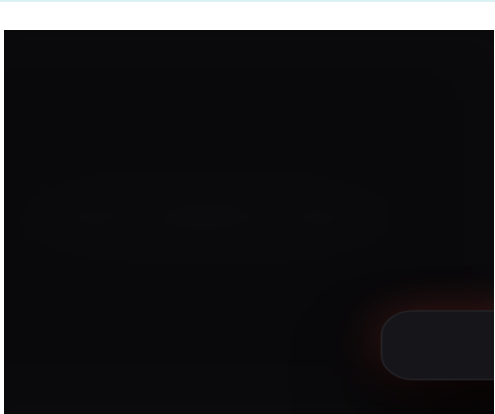
1. RAW STATE

Blur filter & sepia tint



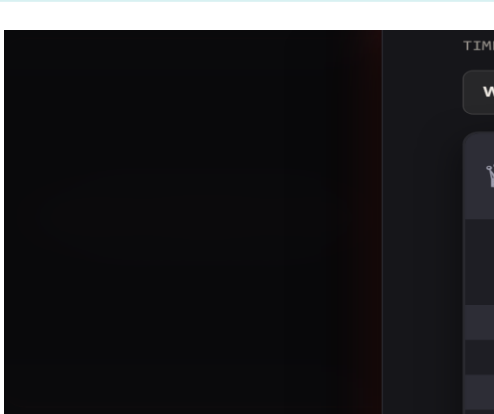
2. HOVER LIT

Safelight glow ignited



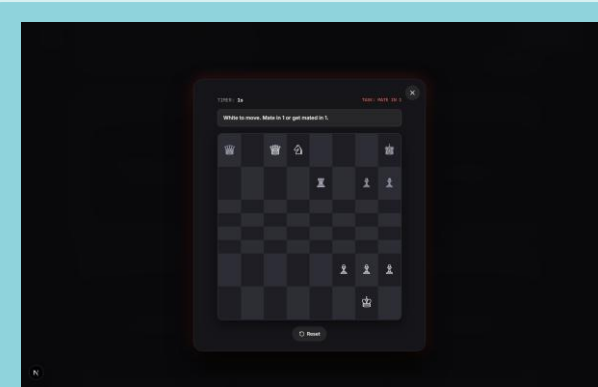
3. MID-DEVELOP

Burst animation transition



4. DEVELOPED

Crisp preview unlocked



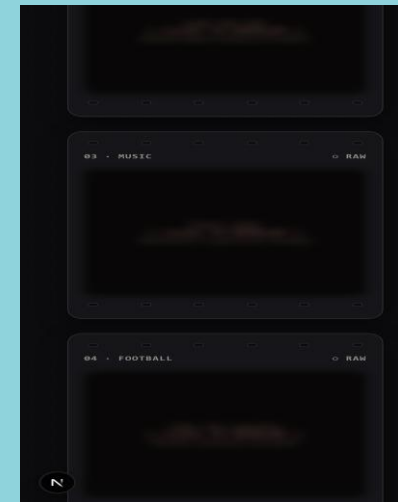
UNLOCKED INTERACTIVE MODAL TOY: Clicking Frame 01 opens a fully playable Chess Tactics puzzle with move timer, FEN parser, and sound cues.

Experience & Motion: Canvas Film Scrubber

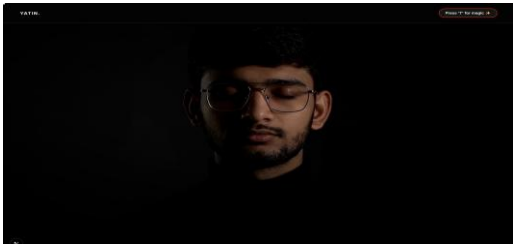


CANVAS SCROLL SCRUBBING (240 FRAMES)

- 240 high-resolution JPEG frames preloaded in controlled parallel batches.
- Autoplay intro loops frames 1..120 at 24 FPS before user scroll takeover.
- Scroll position ratio maps dynamically to current frame via requestAnimationFrame.
- Fallback logic draws nearest loaded frame during fast scrolling.



Anatomy of the Site: Elements & Interactions



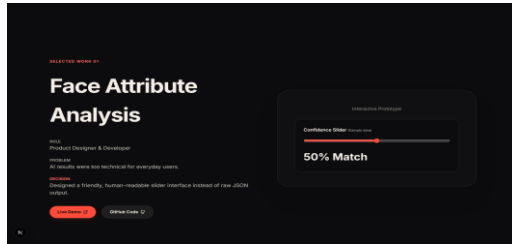
01 · Loading Screen

auto-play preloader



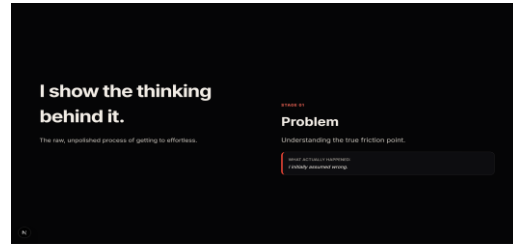
02 · Hero Film Canvas

scroll plays the film



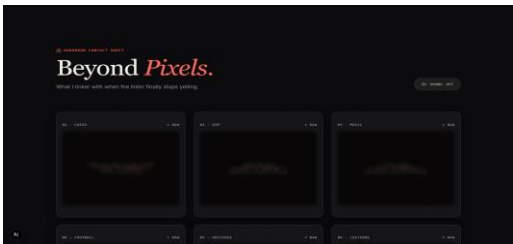
03 · Selected Work

click interactive prototype



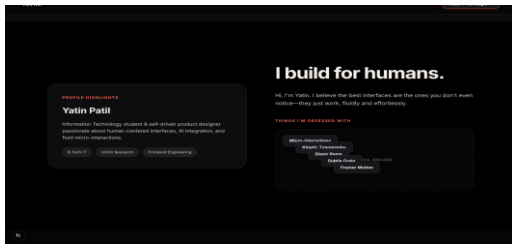
04 · Engineering Process

hover to highlight step



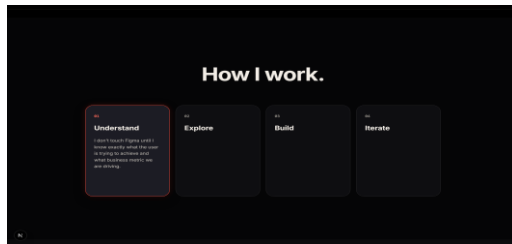
05 · Beyond Pixels

hover dims & click opens toy



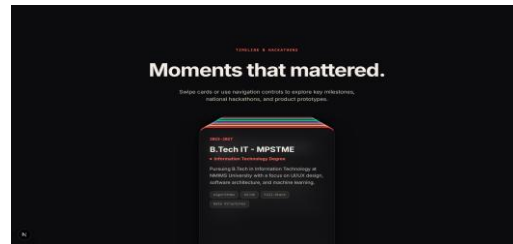
06 · About & Obsessions

hover pills to highlight



07 · Principles Overview

hover principle cards



08 · Moments Deck

click to cycle stack

CUSTOM CURSOR

THEMES WITH T

SOUND OFF DEFAULT

KEYBOARD ARIA

REDUCED MOTION

Design System: Tokens, Component States & Themes

ZONE 1 · VERIFIED TOKENS

-  Base (#0C0C0E)
-  Surface (rgba(18,18,22))
-  Text (#F3EEE7)
-  Muted (#9A958E)
-  Safelight (#FF4B3A)

TYPOGRAPHY ROLES (Inter)

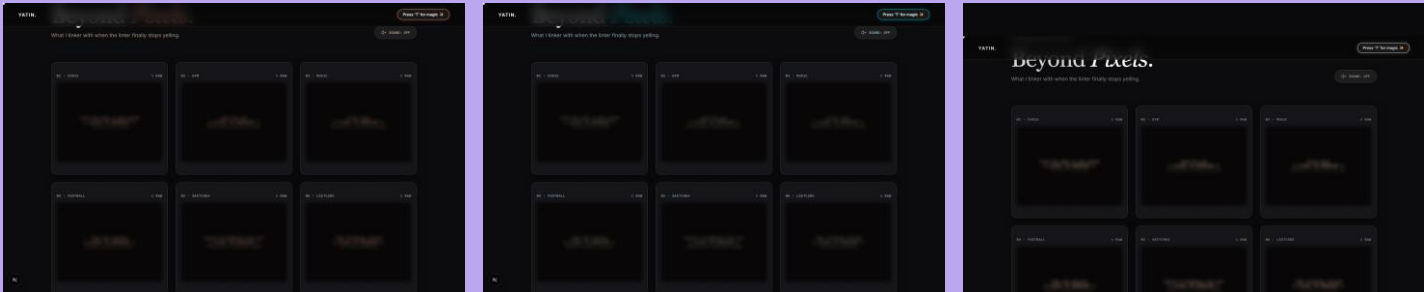
- Display / H1: Inter 700 Bold
- Section H2: Inter 700 Bold
- Body: Inter 400 Regular
- Label: Inter 600 Monospace/Caps

ZONE 2 · COMPONENT STATES



Left: Cursor Dot → Ring → Label State (VIEW/DRAW) | Right: Frame Raw → Lit → Developed

ZONE 3 · ONE LAYOUT, THREE FILM STOCKS (PRESS T TO CYCLE)



1. Portra (warm peach)

2. CineStill (teal)

3. Tri-X (monochrome)

Engineering Lessons: What Went Wrong & Implemented Fixes

PROBLEM: Canvas Frame Lag on Mobile

FIX: Implemented progressive 2-stage chunk loading (frames 1..120 preloaded, rest background pooled).

LESSON: Never block UI threads with synchronous frame fetching.

PROBLEM: Accidental Keyboard Theme Triggers

FIX: Added focus guards for onKeyDown to bypass T-key trigger inside input & editable tags.

LESSON: Keyboard shortcuts must respect DOM focus context.

PROBLEM: Hydration Mismatches in Theme Storage

FIX: Deferred localStorage theme read to useEffect client mount phase.

LESSON: SSR output must remain 100% deterministic during render.

PROBLEM: Accessibility & Reduced Motion

FIX: Integrated prefers-reduced-motion queries to skip canvas autoplay.

LESSON: Motion must be decorative, never mandatory for task completion.

CONCLUSION & FUTURE ROADMAP

Building "The Darkroom" proved that modern portfolios can combine high performance with rich tactile personality.

WHAT'S NEXT (FUTURE ROADMAP):

- Mobile Frame Pool: Serving smaller 720p webp frames on mobile connections.
- Direct Skip Intro: Adding explicit intro skip button for returning visitors.
- WebGL Shaders: Upgrading darkroom developer liquid blur with custom GLSL shaders.
- Interactive Claims Ledger: Live verification panel for all software claims.

EXPLORE THE LIVE PORTFOLIO



<https://yatinportfolio.vercel.app>